

FIFTH SEMESTER DIPLOMA EXAMINATION IN CIVIL ENGINEERING

MODEL QUESTION PAPER

GEOTECHNICAL ENGINEERING

(Maximum marks: 100)

(Time: 3hrs)

Part - A

(Answer all questions)

[Maximum marks -5x 2 =10 marks]

I. Answer the following questions in one or two sentences

1. Define void ratio of soil.
2. Define compaction.
3. Differentiate ultimate and safe bearing capacities of soil.
4. Differentiate shallow and deep foundations.
5. Define the significant depth for foundation analysis.

PART – B

(Maximum marks – 5x6 =30 marks)

(Answer any five questions)

- II.
1. Establish the relationship between porosity and void ratio .
 2. Explain hydrometer method for particle size analysis.
 3. Explain compaction curve and OMC.
 4. state Darcy's law and mention its application .
 5. Explain wash boring method for soil investigation.
 6. What are the steps involved in erection of well foundation.
 7. Give the types of deep foundations with their significance.

PART – C

(Answer any one full question from each module)

(Maximum marks – 4x15=60)

Module – I

- III.
- a. A sample of soil has a volume of 60 ml. and mass of 100gm. After oven drying the mass reduced to 88gm. Find its water content, void ratio, porosity and dry density assuming specific gravity of soil solids as 2.5 (7)
 - b. Give procedure for hydrometer method of particle size analysis. (8)

OR

- IV. a. Define the terms i) bulk density. ii) Dry density. iii) Specific gravity.
iv) Degree of saturation v) Air content vi) Thixotropy (8)
b. Define Atterberg's limits (7)

Module – 2

- V. a. Give classification of soil water. (5)
b. List down the factors affecting permeability of soil. (5)
c. Explain any two field methods for compaction. (5)

OR

- VI. a. Explain the factors affecting permeability of soil. (8)
b. Distinguish discharge velocity and seepage velocity. (7)

Module – 3

- VII. a. What are the different stages of soil investigation? (5)
b. Explain the procedure of plate load test for bearing capacity of soil. (10)

OR

- VIII. a. Explain the procedure of conducting seismic method for soil investigation (8)
b. List down the various methods for soil exploration. (7)

Module – IV

- IX. a. What are the objectives of foundation? (5)
b. Design a column footing subjected to total load of 300KN founded on soil with safe bearing capacity 150 KN/m^2 . (5)
c. Explain any two types of pile foundations with the help of sketches. (5)

OR

- X. a. Mention the conditions in which a strap footing is given for foundation. (5)
b. Sketch and explain well foundation in detail. (10)